SMARTBRIDGE
Let's Bridge the Gap

Project Initialization and Planning Phase

Date	15 March 2024
Team ID	
Project Title	EV Market Performance and Insights
Maximum Marks	3 Marks

Project Proposal (Proposed Solution) template

This project proposal outlines a solution to address a specific problem. With a clear objective, defined scope, and a concise problem statement, the proposed solution details the approach, key features, and resource requirements, including hardware, software, and personnel.

Project Overview	
Objective	To build interactive Tableau dashboards that give EV market stakeholders a clear, data-driven view of sales, manufacturers, battery technology, driving range, customer ratings and environmental impact, so they can plan products, markets and investments with confidence.
Scope	Analysis of the EV market dataset across vehicle categories, manufacturers, manufacturing countries, battery types (LFP, NMC, NCA, Sodium-Ion, Solid-State) and market regions for model years 2017-2027. Delivers two Tableau dashboards for executives, business intelligence managers and researchers. Out of scope: real-time data feeds and predictive models.
Problem Statement	
Description	EV market data (sales, price, battery specifications, range, charging, manufacturing, customer ratings and CO2 savings) is large and scattered, so executives, BI managers and researchers struggle to see which vehicle categories, manufacturers and battery technologies perform best, and where growth and environmental benefit are highest.
Impact	Solving this gives decision-makers one reliable view of the market. It speeds up decisions, improves product positioning and market-entry planning, helps identify high-growth segments and leading battery technologies, and supports evidence-based policy for reducing carbon emissions.
Proposed Solution	

Approach	Connect Tableau to the EV dataset, clean and prepare the data, create calculated fields for the KPIs (total sales, average price, average range, CO2 saved, retained value), build a worksheet for each business question, and combine them into two interactive dashboards with a shared vehicle-category legend, filters and tooltips.
Key Features	KPI cards (total EV sales 1.55B, average price $64,921, average range 324.54 km, CO2 saved 738K tons per year); production output by country and category; top manufacturers by customer review vs sales units; driving range distribution; 3-year retained value; sales trend 2017-2027; battery technology performance; charging vs efficiency; regional CO2 saved by drivetrain.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Standard laptop or desktop with an Intel Core i5 (or equivalent) processor, 4+ cores; no GPU required
Memory	RAM specifications	8 GB minimum (16 GB recommended for large extracts)
Storage	Disk space for data, models, and logs	At least 5 GB free SSD space for the dataset, Tableau extracts and workbooks
Software		
Frameworks	Python frameworks	Tableau Desktop for analysis and dashboards; Tableau Public for sharing
Libraries	Additional libraries	Tableau built-in chart types (bar, line, box plot, scatter, bubble, dual-axis, stacked area), calculated fields and LOD expressions

Development Environment	IDE, version control	Tableau Desktop and Microsoft Excel; workbook versions saved as .twb / .twbx files with a cloud backup
Data		
Data	Source, size, format	EV market dataset (CSV / Excel) with sales, price, battery, range, charging, efficiency, manufacturer, country, rating and CO2 fields for model years 20172027